



140A 100V Dual Field-Oriented-Control Brushless Motor Controller

1 Features

- Dual fully independent Field-Oriented-Control for brushless motors.
- Maximum input voltage of 100-V and 140-A continuous per motor with adequate cooling
- Supports multitude of control modes including current, duty cycle, speed and position control (position control works best with encoder)
- Open-Source firmware/scripting supports coding of custom applications that can run on the onboard STM32F405 MCU.
- 6 low-side shunts and amplifiers with careful signal processing for precise phase current measurements
- Configurable sensor input ports compatible with hall sensors, ABI encoders, and some SPI based magnetic encoders
- Broad range of communication interfaces including USB, Bluetooth, Wi-Fi, CAN, SPI, I2C, UART, PWM, and ADC inputs
- Integrated IMU with 3-Axis Accelerometer and 3-axis Gyroscope
- Fully configurable sleep modes with ultra low sleep current to support under-voltage-lockout, and auto power-up and shutdown
- Integrated temperature sensors automatically protect the system during high power draw
- Easy setup with VESC TOOL Desktop and Phone applications for fast integration alongside the possibility for deep customization
- Integrated ESP32-C3 supports both station and access-point Wi-Fi connections alongside BLE 5
- Integrated local storage supports data-logging and optional GPS module can be added for geo-tracking
- Switched 12V and Bus Voltage outputs for auxiliary system devices

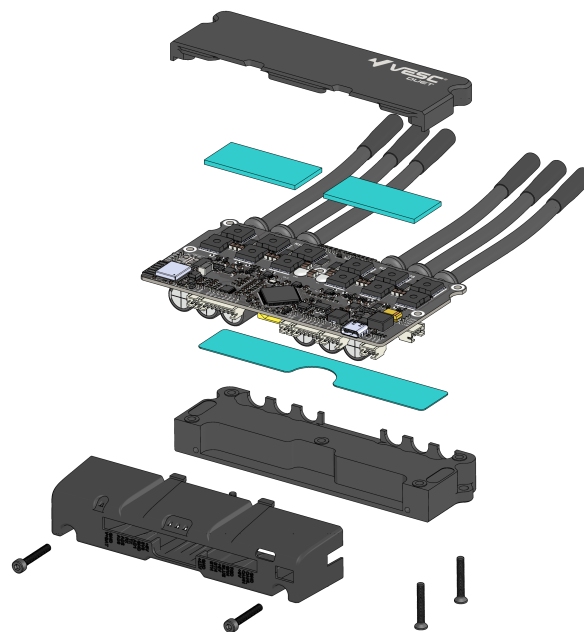


Figure 1: An exploded view of the VESC Duet

2 Description

The VESC Duet is a feature-rich dual brushless motor controller which can provide up to 8-kW of continuous power across two motors when cooled appropriately. An on-board STM32F405 micro-controller running open-source VESC software enables precise independent actuation of two motors alongside configurable custom applications to control them. The open-source firmware comes packaged with several ready-to-use applications to address common implementations of the hardware for electric mobility and automation. It's breadth of external communication interfaces enables easy integration with many different systems and peripherals. It supports both sensorless and sensed brushless motor applications with full sinusoidal Field-Oriented-Control enabling quiet, efficient, and powerful motor dynamics. It also supports other advanced control techniques including overmodulation and field weakening for maximizing power and speed in certain applications, and high frequency injection for smooth sensorless low-speed control.

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3 Specifications

3.1 Absolute Maximum Ratings

Parameter	Min	Max	Unit
Input Voltage	15	100 ¹	V
Battery Series (Li-Ion)	6	22	S
Continuous Input Current ²		140	A
Pulsed Input Current ²		200	A
Continuous Motor Current (per motor) ³		140	A
Pulsed Motor Current (per motor) ³		200	A
PWM Switching Frequency		25	kHz
5-V Auxiliary Current		1	A
3.3-V Auxiliary Current		0.5	A
+12V 2-pin port current draw		0.8	A
Input Voltage 2-pin port current draw		0.8	A
Servo ppm input	2	5.5	V
ADC inputs	0	3.3	V

3.2 Recommended Operating Conditions

Parameter	Min	Max	Unit
Input Voltage	12	85	V
Motor Current (per motor)	-140	140	A
V+ Switched Output Current	0	0.5	A
Servo ppm input	3	5	V
ADC inputs	0	2.8	V
Operating Temperature	-40	85	°C
Humidity	Non-condensing		-

3.3 Typical Operating Characteristics

Parameter	Typ.	Max	Unit
12V Bus Voltage	11.8	12.3	V
Sleep Input Current (72V Input)	3	20	uA
Awake resting Input Current (72V Input)	5	30	mA

¹Spikes may not exceed this, some margin should be accounted for. When using li-ion cells that charge up to 4.2V per cell at most 22 cells in series are recommended.

²The input current listed refers to the total current running through the power input connector of the ESC. Note that in the VESC Tool settings the total input current will be the sum of each individual motors set input current, so for balanced power delivery to both motors one half of this total current should be set for each motor.

³The motor current listed is referring to the absolute current in the Dq frame of the motor and thus is related to the peak amplitude current any one motor phase wire will experience. A stalled motor running against the current limit can experience the peak current continuously if it happens to stall when aligned directly with a given phase.

3.4 Physical Properties

Parameter	Value	Unit
Length	134	mm
Width	70	mm
Height	24	mm
Mass	418	g

4 Pinout and Wiring

4.1 Pin Descriptions

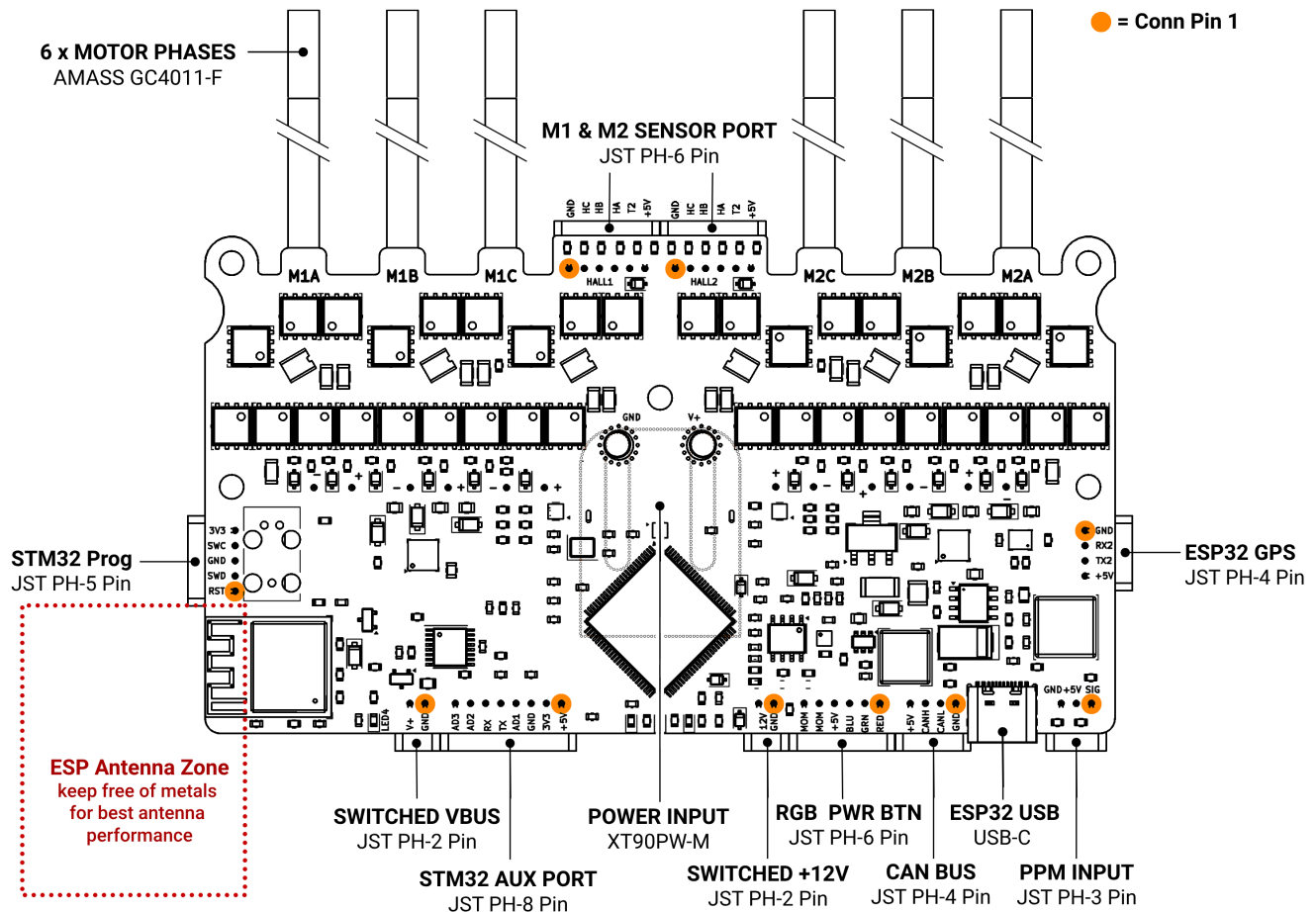


Figure 2: Top down view of Duet Circuit Board which is how it is oriented when looking down at a controller resting face up on a table in its housing. Bold text refers to the connector names as detailed in the table below and undertext references the connector part number to assist in sourcing mating connectors. The orange circles define pin 1 of each connector as defined by the manufacturer.

Pin #	Pin Name	Description
Motor Phases		
1-6	M1A-M2C	Phase connections for motor leads. They each have a 130mm 12AWG wire with 4mm GC4011-F female bullet from AMASS affixed. For larger batch orders this can be customized by contacting us.
Power Input		
1	VBUS	Connection to positive power input. This pin connects directly to the bulk capacitors (~2 milliFarad) so sparks can be generated if an XT90-S (antispark) connector is not used.
2	GND	Connection to ground power input. This controller operates on common ground but the low side of the power input (battery or DC supply) to the controller should be connected here.

M1 &M2 Sensor Ports		
1	GND	Connected to ground
2 - 4	HALL A,B, C	Hall input connections, see fig. 3 for details.
5	MOT TEMP	Motor temperature sensor input for 10 kOhm 3380 beta NTC thermistor which should be connected between this pin and ground. See fig. 3 for details.
6	5V	Connects through schottkey diode to internal 5 V bus. Pin should only be used to source less than 0.2 Amps and will see a forward voltage drop between 0.3-0.5V leaving the output between 4.5-4.7 volts.
ESP32 GPS		
1	GND	Ground reference pin
2	RX2	Recieve data pin connected to GPIO20 of the ESP32-C3. Should be connected to TX of GPS module.
3	TX2	Transmit data pin connected to GPIO21 of the ESP32-C3. Should be connected to the RX pin of GPS module.
4	+5V	5V output for GPS module
PPM INPUT		
1	GND	Ground reference pin
2	5V	5V output for Remote Receiver
3	SIG	PPM Signal input to MCU, has an analog lowpass filter with 7 kHz cutoff frequency
ESP32 USB		
USB-C connection to ESP32-C3. Can be used to connect to STM32 through CAN forwarding in VESC Tool. +5V is left unconnected through an unpopulated resistor so externally powering the duet is required.		
CAN BUS		
1	GND	Ground reference pin, do not connect if devices are already sharing common ground from same power source
2	CANL	CAN bus low pin for differential CAN data
3	CANH	CAN bus high pin for differential CAN data
4	+5V	5 volt output, do not connect if other device is powered independently
Pin #	Pin Name	Description
RGB PWR BTN		
1 - 3	RED GRN BLU	Connected through 220 Ohm Resistor to Pin PD11, PD12, and PD15 of the STM32, respectively. Used by default to sink current from the cathode of a rer, green, or blue LED in the included RGB switch.
4	+5V	5V source pin for common anode of RGB LED
5	MOM2	Input to discrete logic for latching enable pin of onboard 12V regulator. Shorting this pin to MOM1 for a set time will latch the unit on until a configurable period of inactivity when the duet will return to sleep.
6	MOM1	Connects a 100 kOhm pull-up resistor to the input voltage
Switched +12V		
1	GND	Ground reference pin

2	12V	Switched 12V output with short circuit protection. Maximum source current of 720 mA. Can be controlled using the auxiliary output mode setting in VESC Tool from the Motor Settings →General →Advanced-tab. Is also possible to control this output from LispBM using the (set-aux 1 x) extension.
STM32 AUX PORT		
1	+5V	5 volt auxiliary power pin
2	3V3	3.3 volt auxiliary power pin
3	GND	Ground reference pin
4, 7, 8	AD1-3	Analog inputs for throttle, break, and other signals. Each has an analog RC filter in series see fig. 3 for details.
5	TX	transmit pin for uart communication, clock (SCL) pin for i2c
6	RX	receive pin for uart communication, data (SDA) pin for i2c
Switched VBUS		
1	GND	Connected to ground
2	V+	High-side switched bus voltage output capable of sourcing up to 1 amp. Switched with optocoupled mosfet by STM32 and has a surface mounted 1 amp fast blow fuse for added protection of the rest of unit. Has a catch diode to prevent reverse voltage spikes but should not be used to switch large capacitive loads since it has no precharge circuitry. This output is mapped to AUX 2 on the STM32 and can be controlled from LispBM using the (set-aux 2 x) extension.
STM32 Prog		
1	RST	Reset pin for STM32, pull low to reset
2	SWD	Data pin for STM32 programming port
3	GND	Ground reference pin
4	SWC	Clock pin for STM32 programming port
5	3V3	3.3V bus connection

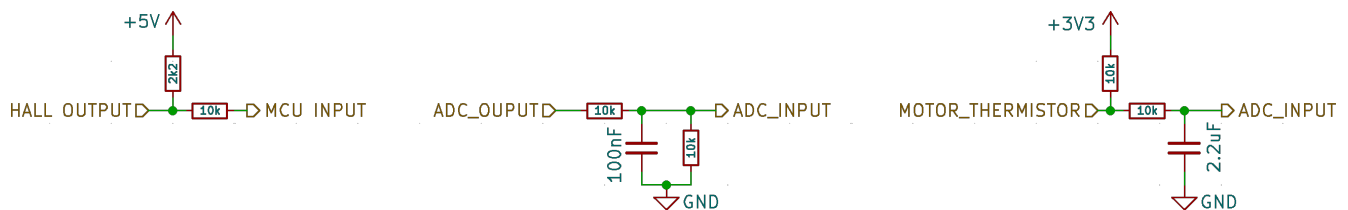


Figure 3: Internal filtering and protection diagram for external motor hall sensor inputs, external ADC inputs such as analog throttle and break modules, and Motor Temp Sensor inputs, respectively. The hall inputs have a 2.2 kOhm pull-up resistor to +5V and a 10 kOhm resistor to prevent damage to the MCU in the event of short circuits to voltages in motor. The external ADC inputs have a similar series resistor and a 160 Hz low-pass filter to attenuate noise alongside a 10 kOhm pulldown to voltage divide 5 volt input signals to prevent saturation and to allow detection of broken or disconnected throttle and break wires so they can be handled appropriately. The motor thermistor input has a pullup to 3.3V so a 10k 3380 beta value NTC thermistor can be connected between this pin and ground. A 7 Hz low pass filter in series smooths any noise on this measurement and prevents damage to the STM32

4.2 Wiring Diagram

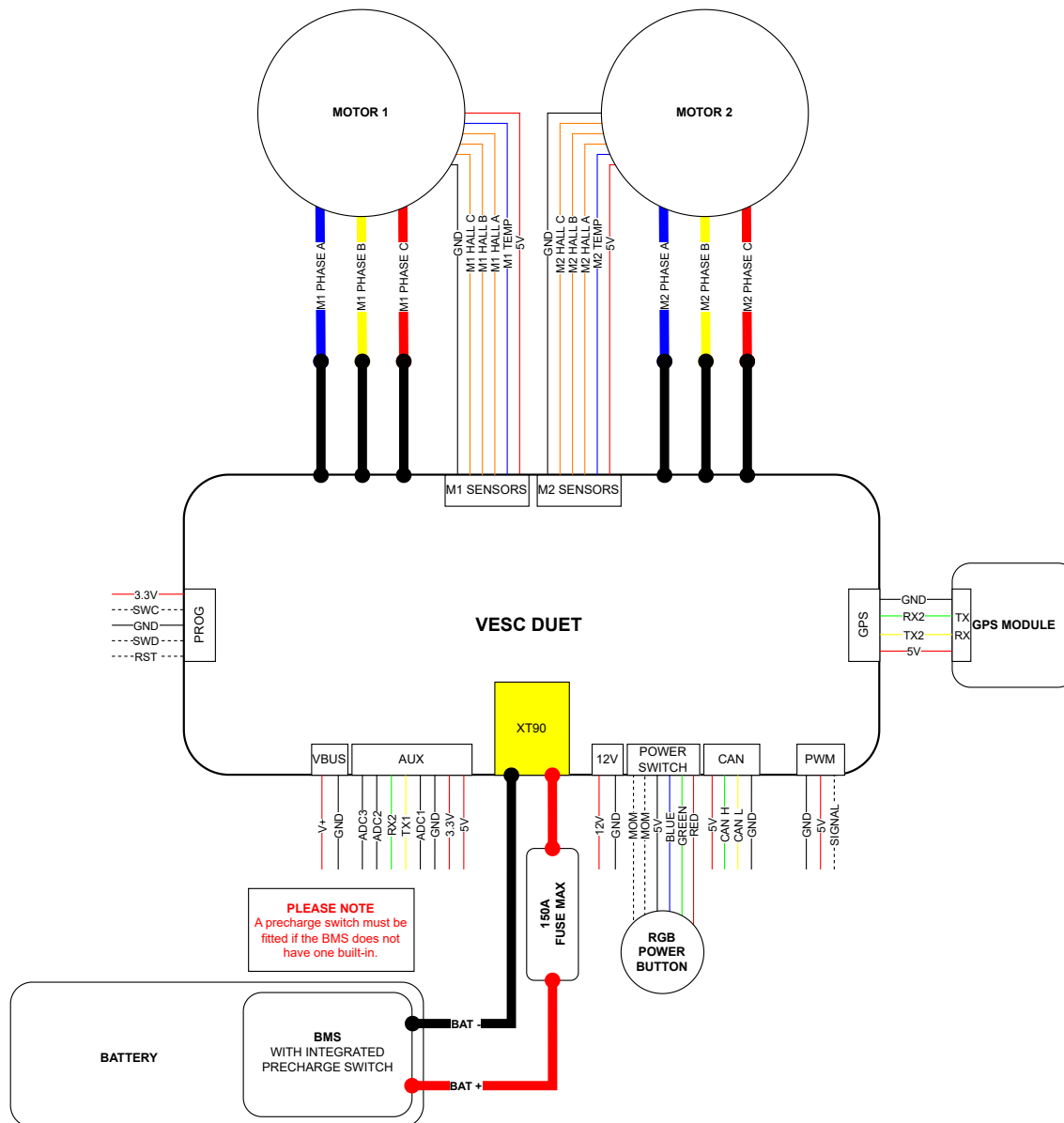


Figure 4: Composite image of VESC Duet with two motors, battery and BMS with precharge switch.

Please note: An XT90-S (antispark) connector can be used as a precharge switch on this device. Also note that XT90-S have a limited number of cycles. If hear a pop or a spark please replace the XT90-S.

5 Device Installation

5.1 Installation Warnings

- **Installation must be performed by qualified personnel** with experience in high-voltage battery systems and VESC-based motor controllers.
- **Read and fully understand all instructions, datasheets, and safety warnings** before beginning installation.
- **Gather all necessary tools and materials** including the device, wiring harnesses, connectors, insulation materials, and personal protective equipment.
- **Ensure the battery pack and motor system are prepared and properly arranged** with correct wiring paths and secure mounting.
- **Securely mount the device using threadlock** in a location protected from vibration, moisture, dust, and direct heat sources.
- **Do not install the device in locations exposed to water, condensation, or flammable materials.**
- **Install motor temperature sensors as required**, in order for the motor protection to ramp down the current smoothly when the motor is about to overheat.
- **Only use manufacturer-specified connectors and cables** to ensure safe and reliable connections.
- **Verify correct polarity for all connections.** Incorrect wiring can cause irreversible damage to the device, battery pack, or connected equipment. **Double-check the wiring diagram before proceeding.**
- **Insulate all exposed terminals and wiring** to prevent accidental short circuits or electric shock.
- **Keep metal objects and tools away from the device and power terminals during installation** to avoid accidental bridging of electrical contacts.
- **Tidy up and secure all cables** with cable ties or routing guides to prevent movement and wear.
- **Do not drill, cut, or modify the device enclosure or circuit board.** Unauthorized modifications may void the warranty and compromise safety or functionality.
- **Double-check all connections and wiring against the provided diagrams** before applying power to the system.
- **When powering up the system for the first time, do so in a controlled environment with appropriate protective equipment.** Be prepared to disconnect power immediately if any abnormal behaviour is observed.

5.2 Mechanical Dimensions

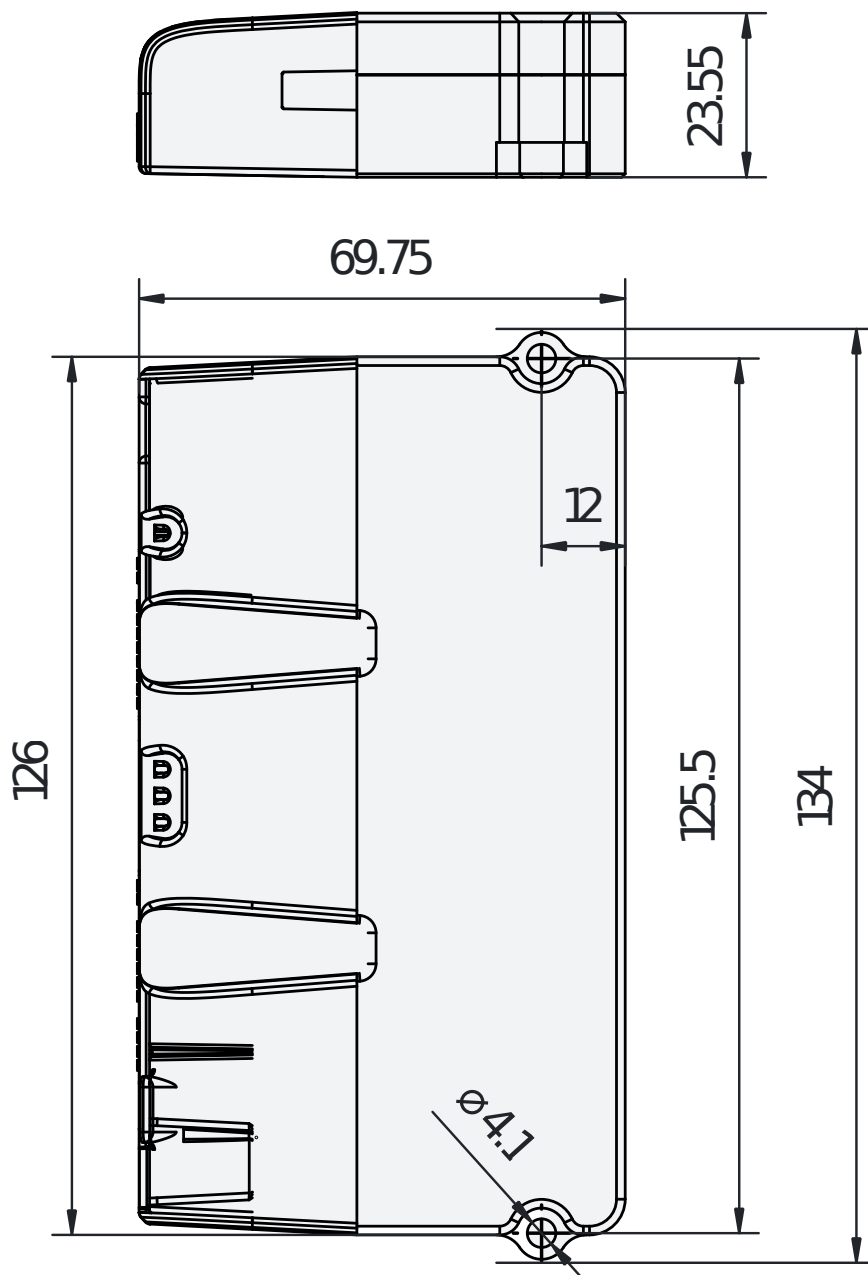


Figure 5: VESC Duet Dimensions shown in mm

6 Configuring the Device

6.1 Configuration Warnings

- **Double-check all wiring connections** before powering on the device. Incorrect wiring can cause damage to the device, battery cells, or pose a fire risk.
- **Always install the latest version of VESC Tool from the official website** to ensure compatibility with your hardware and access to the newest features, bug fixes, and safety updates. Download the installer from vesc-project.com/vesc_tool and follow the on-screen instructions. After installation, update your device firmware if prompted to ensure optimal performance.
- **Do not exceed the recommended voltage limits** for this device, your specific battery or motor system. Setting incorrect voltage thresholds can result in overvoltage errors or undervoltage cutoffs, potentially damaging the system.
- **Do not exceed the recommended current limits** for this device, your specific battery or motor system. Setting incorrect current thresholds can result in overheating or excessive stress on components, which may lead to failure or dangerous conditions.
- **Ensure the motor and any moving parts connected to it are free from obstruction and lifted off the ground before powering on the device.** This prevents unexpected movement that could cause injury, damage, or interference with calibration and configuration procedures
- **Avoid configuring the device in wet or humid environments.** Moisture can cause short circuits and damage sensitive electronic components.
- **If you are unsure about any configuration step, consult the datasheet or seek professional assistance.** Improper configuration can void warranties and lead to hazardous conditions.
- **Wear appropriate personal protective equipment (PPE),** such as safety glasses and gloves, when working with high-voltage systems.

6.2 Connecting the Controller to VESC Tool

The controller can be connected to VESC Tool via USB, Wi-Fi, Bluetooth, or CAN. For instructions on how to connect with and navigate VESC Tool please visit <https://www.vesclabs.com/category/getting-started/>.

1. Connect the controller to VESC Tool using one of the mentioned methods.
2. Ensure the **“VESC Duet”** has been selected from the list of CAN devices if you have connected through another device via CAN.

6.3 Using the Motor Setup Wizard

1. On the **Welcome & Wizards** page select the **“Setup Motors FOC”** wizard.
2. Confirm that you want to load default parameters by pressing **“Yes”**.
3. Select the usage which matches closest to your application followed by pressing **“Next”**.
4. Select the size of motor which is connected to the controller followed by pressing **“Next”**.
5. Select the **“Battery Type”** from the drop-down list.
6. Enter the number of **“Battery Cell Series”**.
7. Enter the **“Battery Capacity”** followed by pressing **“Next”**.
8. Select whether the system is direct drive or not.
9. Enter the values of **“Motor Pulley”** and **“Wheel Pulley”** in order to calculate the gear ratio of your system.

10. Enter the number of **“Motor Poles”** which your motor has.
11. Press the **“Run Detection”** button to continue.
12. **Ensure the motor and any moving parts connected to it are free from obstruction and lifted off the ground before running the detection.** This prevents unexpected movement that could cause injury, damage, or interference with calibration and configuration procedures.
13. Confirm whether you want to **detect the motor** connected to the controller or **all of the motors** connected to the controllers via CAN Bus, followed by pressing **“Ok”** to start the detection.
14. Check the detection results are correct for you system followed by pressing **“Continue”**.
15. Check the **motor direction** is correct for you system by pressing the **“FWD”** or **“REV”** buttons. **If the direction is incorrect**, select the **“Inverted”** toggle button to swap the direction of the motor.

6.4 Configuring Motor Current

1. Navigate to the **“General Motor Settings”** section from the menu on the left of the screen (on desktop), or by swiping across the screen on a mobile device.
2. Select the **“Current”** tab at the top of the screen.
Before proceeding check the limits of your controller and motor.
Do not exceed the current limits of the controller or motor!
3. Enter the value of **“Motor Current Max”** rated for your motor or application.
4. Enter the value of **“Motor Current Max Brake”** rated for your motor or application.
5. Enter the value of **“Absolute Maximum Current”** rated for the motor or application.
6. Press the **“Write Motor Configuration”** button to write the new configuration to the controller.
7. If **multiple controllers** have been connected via CAN (or a **dual controller**), swap CAN devices and follow steps 1-6 to apply the settings to the other controllers connected.

6.5 Configuring Battery Current

1. Navigate to the **“General Motor Settings”** section from the menu on the left of the screen (on desktop), or by swiping across the screen on a mobile device.
2. Select the **“Current”** tab at the top of the screen.
3. Enter the value of **“Battery Current Max”** rated for your battery or application.
Battery current max can be calculated from the **maximum output current** of your battery or BMS, divided by how many motors are in the system.
4. Enter the value of **“Battery Current Max Regen”** rated for your battery or application.
Battery current max regen can be calculated from the **maximum charge current** of your battery or BMS, divided by how many motors are in the system.
5. If multiple controllers have been connected via CAN (or a **dual controller**), swap CAN devices and follow steps 1-4 to apply the settings to the other controllers connected.

6.6 Configuring Inputs

1. On the **Welcome & Wizards** page select the **“Setup Input”** wizard.
2. Press the **“Next”** button which will start scanning for other devices connected via CAN bus.
3. Select the type of input you would like to control the controller.
4. Move the input(s) to the maximum position, including forwards, backwards and centre on bi-directional inputs.
5. Press the **“Apply”** button to map the input.
6. Move the input(s) to the maximum position, checking that the input has been mapped correctly.
7. Select the **“Next”** button to continue.
8. Select the control type for your application from the drop-down list. Please use the “?” icon which explains the difference in control types.
9. Select the **“Next”** button to continue.
10. Select the **“Finish”** button to complete the setup.

7 Power Switch

Included with the Duet is a RGB power switch. This switch not only functions as a method of turning the device on and off but also gives an indication of the batteries state of charge and whether the device has a fault. See the fault code section for more details.

7.1 Powering the Device ON or OFF

To turn the device **ON**, press the power button. The switch will then show a rainbow animation, indicating the device has been powered on before changing to a colour indicating the batteries state of charge.

To turn the device **OFF**, hold down the power button for 3 seconds. The switch will show no colour and therefore the device has been powered off.

7.2 Battery State of Charge

Once the device is powered on the colour of the power switch indicates the batteries state of charge and will fade from green to red as the battery is depleted. See below for examples:



Figure 6: Power switch indicating 0% battery - RED



Figure 7: Power switch indicating 50% battery - ORANGE



Figure 8: Power switch indicating 100% battery - GREEN

7.3 Fault Code Indication

If the device has a fault the power switch will flash RED then pause. The number of flashes corresponds to the fault code. See the fault code section for more information.

8 Fault Codes

Fault codes can be seen by connecting the device through the VESC Tool.

1. Go to the **“Terminal”** tab at the top of the screen and click **“Print Faults”**.
2. Alternatively go to the **“VESC DEV Tool”** section from the menu.
3. Type in the VESC terminal **“faults”**.
4. Press the **“Send Command”** button which will list the faults on the device in the terminal.

See a list of fault codes below:

Number	Fault Code
0	None
1	Over Voltage
2	Under Voltage
3	DRV
4	Absolute Over Current
5	Over Temp FET
6	Over Temp Motor
7	Gate Driver Over Voltage
8	Gate Driver Under Voltage
9	MCU Under Voltage
10	Booting from Watchdog Reset
11	Encoder SPI
12	Encoder Sincos Below Min Amplitude
13	Encoder Sincos Above Max Amplitude
14	Flash Corruption
15	High Offset Current Sensor 1
16	High Offset Current Sensor 2
17	High Offset Current Sensor 3
18	Unbalanced Currents
19	BRK
20	Resolver LOT
21	Resolver DOS
22	Resolver LOS
23	Flash Corruption App Config
24	Flash Corruption MC Config
25	Encoder No Magnet
26	Encoder Magnet Too Strong
27	Phase Filter
28	Encoder Fault
29	LV Output Fault

9 Product Warranty

9.1 Warranty Coverage

Please find the warranty information regarding VESC Labs products using the following link:

<https://www.vesclabs.com/warranty-and-returns-policy/>

10 Compliance & Regulatory Information

10.1 CE Marking

This product meets the requirements of all applicable EU directives and regulations and complies with the essential health, safety, and environmental protection requirements as set out in relevant legislation. It is CE marked in accordance with:

- **Directive 2014/30/EU** – Electromagnetic Compatibility (EMC)
- **Directive 2014/35/EU** – Low Voltage Directive (LVD)
- **Directive 2011/65/EU** – Restriction of Hazardous Substances (RoHS)

10.2 UKCA Marking

This product is compliant with UK regulations and bears the UKCA marking, confirming its conformity with the applicable statutory requirements in Great Britain. It meets the relevant provisions of:

- Electrical Equipment (Safety) Regulations 2016
- Electromagnetic Compatibility Regulations 2016
- The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012

10.3 WEEE Directive (2012/19/EU)

This product is compliant with the Waste Electrical and Electronic Equipment (WEEE) Directive. It should not be disposed of with unsorted municipal waste and must be collected separately for proper treatment and recycling. Please contact your local distributor or relevant waste authority for information on recycling options and collection facilities.

